

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	KS THARUNESHWAR	Reg. No.:	192572148
Section / Batch:	2025	Date:	20 . 07. 2026

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type		Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2, Q3	40 60 (2X30)
GRAND TOTAL:		100 Marks	

SECTION A - APPLICATION BASED QUESTIONS

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Total Comparisons and Swaps

Pass	Comparisons	Swaps
1	5	5
2	4	3
3	3	1
4	2	1
5	1	1
Total	15	11

Total comparisons : $15 = \frac{n(n-1)}{2}$ [W.K.T, $n=6$, $\frac{6(5)}{2}$]
 $= 15$]

Total swaps : 11 (out of maximum possible 15)

Time Complexity Analysis:

Best Case:

→ Array already sorted

→ Complexity : $O(n)$

Average Case:

→ Elements in random order

→ $O(n^2)$

Worst Case:

→ Array sorted in reverse order

→ $O(n^2)$

Space Complexity: $O(1)$

5) Verification of Final Output:

Final sorted array: $[5, 10, 20, 30, 40, 50]$

Checking: $5 < 10 < 20 < 30 < 40 < 50$

Correctly sorted in ascending order

6) Efficiency Interpretation on Alternating Values:

An alternating sequence like $[50, 10, 40, 20, 30, 5]$ is inefficient for bubble sort because many adjacent elements are out of order, resulting in frequent swaps (11 out of 15).

Since bubble sort moves elements only one position at a time, values far from their correct position require multiple passes, giving it an $O(n^2)$ time complexity on disordered data.

Code of Conduct (192525213)

I, **KS THARUNESHWAR 192572148** (Reg No) certify that this submission is my original work and that I have assessed. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **KS THARUNESHWAR**

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
(40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
(30 Marks)	4	3	3	2	2	22
Overall Total 100						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____